

1. Open your WSL Ubuntu terminal

Open Windows Terminal/PowerShell and enter:

➤ `Wsl`

Then switch to the Hadoop user if necessary:

➤ `su - Hadoop`

Check:

➤ `echo $HADOOP_HOME`

2. Check that Hadoop is actually working

Run:

➤ `jps`

You should eventually see processes such as:

NameNode
DataNode
ResourceManager
NodeManager

If they are not running, start Hadoop:

➤ `start-all.sh`

One thing to verify

Run:

➤ `java -version`

and:

➤ `javac -version`

We need both to be Java 8.

If `javac -version` shows Java 25 or other, run:

➤ `sudo update-alternatives --config java`

➤ `sudo update-alternatives --config javac`

and select the Java 8 compiler.

3. Find your Hadoop JARs

This is the most important part for your Eclipse setup.

Run:

```
➤ find $HADOOP_HOME/share/hadoop -name "*.jar" | head -30
```

You should see paths similar to:

```
/home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
```

```
/home/hadoop/hadoop/share/hadoop/common/hadoop-common-3.3.6-  
tests.jar
```

```
/home/hadoop/hadoop/share/hadoop/hdfs/hadoop-hdfs-3.3.6.jar
```

```
/home/hadoop/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-client-  
core-3.3.6.jar
```

```
/home/hadoop/hadoop/share/hadoop/yarn/hadoop-yarn-api-3.3.6.jar
```

...

This is where your Hadoop installation keeps its Java libraries.

4. Create the Hadoop_java_jars folder

We can create dir in WSL.

Run:

```
➤ mkdir -p ~/Desktop/Hadoop_java_jars
```

Check:

```
➤ ls ~/Desktop
```

You should see:

```
Hadoop_java_jars
```

5. Copy Hadoop JARs into that folder

```
➤ find $HADOOP_HOME/share/hadoop -type f -name "*.jar" -exec cp -n {}  
~/Desktop/Hadoop_java_jars/\;
```

6. Installing Eclipse

First go to your home directory:

➤ `cd ~`

Download Eclipse IDE for Java Developers:

➤ `wget -O eclipse.tar.gz`
`https://download.eclipse.org/technology/epp/downloads/release/2024-03/R/eclipse-java-2024-03-R-linux-gtk-x86\_64.tar.gz`

Then extract:

➤ `tar -xzf eclipse.tar.gz`

You should now have:

`~/eclipse`

Check:

➤ `ls ~/eclipse`

You should see something such as:

`eclipse`

`eclipse.ini`

`plugins`

`features`

Start Eclipse

Run:

➤ `~/eclipse/eclipse`

7. Set up Eclipse for Exp 3

Now open Eclipse.

Create:

File

→ New

→ Java Project

Name:

wc1

Then:

Right-click wc1

→ Build Path

→ Configure Build Path

→ Libraries

→ Add External JARs...

Navigate to:

/home/hadoop/Desktop/Hadoop_java_jars

Press:

Ctrl+A

Click Open.

Then:

Apply and Close

You should now see:

wc1

├─ src

├─ JRE System Library

└─ Referenced Libraries

with Hadoop JARs underneath.